# Homework 5

Earth's Atmosphere and Climate
MATH-UA0228

(1) Consider an airplane (or parcel of air) that leaves the equator with zero east-west velocity relative to the surface, that is, $u(0) = 0$, and travels northward towards the pole. Assume that the plane conserves its angular momentum,

$$\text{angular momentum} = m u_a(\phi) r(\phi)$$

where $m$ is its mass, $u_a(\phi) = \Omega r(\phi) + u(\phi)$ is its absolute velocity, which includes that from solid body rotation, $\Omega r(\phi)$, and that relative to the surface, $u(\phi)$. $\phi$ is the latitude.

(a) Compute $u(\phi)$.

(b) In particular, at which latitude does the plane's east-west velocity reach the speed of sound (at surface pressure), 343 m/s.

(c) The fastest "air breathing" air plane (i.e, not a rocket) is the NASA X-43 hypersonic scram jet which goes Mach 9.6, or 3019 m/s. At which latitude does our air parcel's east-west velocity $u(\phi)$ equal that of the X-43?

(2) Obviously air parcels (or our plane) will not travel so fast. For one, friction will slow the plane down, but let's still neglect this (assume there is no viscosity) and focus on another force that will prevent this from happening. The meridional momentum equation with no viscosity is

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z} + fu = -\frac{1}{\rho}\frac{\partial p}{\partial y}$$

where the Coriolis parameter $f = 2\Omega\sin(\phi)$. Assume that $v$ is small, so we can neglect all the nonlinear terms, yielding:

$$\frac{\partial v}{\partial t} + fu \approx -\frac{1}{\rho}\frac{\partial p}{\partial y} = C$$

where the force that moves our plane forward is the pressure gradient term, $\frac{\partial p}{\partial y}$, which is fixed by the strength of our engines: that is, it is equal to some constant, $C$. Explain why our plane cannot keep moving towards the pole and will eventually stop moving northward, with $v = 0$.

(3) By considering the mass of air in a box of width $\delta x$ by $\delta y$ by $\delta z$ centered at the point $(x, y, z)$, derive the conservation of mass equation,

$$\frac{\partial \rho}{\partial t} = -\frac{\partial(\rho u)}{\partial x} - \frac{\partial(\rho v)}{\partial y} - \frac{\partial(\rho w)}{\partial z} = -\nabla \cdot (\rho \mathbf{u})$$

where $\mathbf{u} = (u, v, w)$ is the three dimensional velocity vector.

(4) This question will continue work with conservation of mass equation, help you to brush up on vector notation and make an inference about the divergence of the flow.

(a) Show that $\nabla \cdot (\rho \mathbf{u}) = \rho \nabla \cdot \mathbf{u} + \mathbf{u} \cdot \nabla \rho$

(b) Using the result from your last questions, show that

$$\frac{D\rho}{Dt} = -\rho \nabla \cdot \mathbf{u}$$

(c) If the density does not change much (as with water, or air in geostrophic balance), $\frac{D\rho}{Dt} \approx 0$. Conclude the flow is non-divergent: $\nabla \cdot \mathbf{u} \approx 0$.

We will next consider the zonal momentum equation with viscosity for questions 5–9.

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} + fv = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\frac{\partial^2 u}{\partial x^2}$$

The goal is to non-dimensionalize the momentum equation and find the dominant terms.

(5) First, let's focus on the nonlinear terms,

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z}$$

Assume that $u$ and $v$ have the same scale, $u = U\hat{u}$ and $v = U\hat{v}$, and that they vary over the same length scale, $x = L\hat{x}$ and $y = L\hat{y}$. The vertical velocity is different, though: $w = W\hat{w}$ and $z = H\hat{z}$.

(a) Show that $u\frac{\partial u}{\partial x}$ and $v\frac{\partial u}{\partial y}$ scale as $U^2/L$.

(b) Show that $w\frac{\partial u}{\partial z}$ scales as $UW/H$.

(c) Assume that flow is non-divergent, as justified in problem 4(c), so that

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

Scale this equation and conclude that $U/L \approx W/H$.

(d) Finally, show that all three of the nonlinear terms scale as $U^2/L$.

(6) Now we will compare the nonlinear terms to the strength of the viscosity term.

(a) Show that the viscous term $\nu \frac{\partial^2 u}{\partial x^2}$ scales as $\nu U/L^2$

(b) Based on the last problem, the nonlinear terms scale as $U^2/L$. Show that ratio of the nonlinear terms to the viscosity is

$$\frac{UL}{\nu} \equiv Re$$

This result is so important that we gave it a name, the Reynolds number $Re$.

(c) In the atmosphere, the strength of the wind $U \approx 10$ m/s and the length scale of a storm $L \approx 1000$ km $= 10^6$ m, and the kinematic viscosity of air is approximately $\nu \approx 1 \cdot 10^{-5}$. What is the Reynolds number $Re$ and what can you conclude about the importance of viscosity.

(d) Above you should have concluded that the viscous term is incredibly tiny compared to the nonlinear terms. We will ignore it in the later questions, but something does have to slow down the wind; otherwise it would keep growing! The key is that viscosity acts at smaller length scale $L$. Keeping $U \approx 10$ m/s and $\nu \approx 1 \cdot 10^{-5}$, for what length scale $L$ would the Reynolds number be approximately 1?

(e) This result shows that it is impossible to simulate the atmosphere with the molecular viscosity. You would need a grid that resolves the length scale you computed above, but is large enough to capture the whole world! To get a sense of how massive this would be, let's not worry about spherical geometry, but just imagine you want to simulate the wind in a box that is $10^7$ by $10^7$ by $10^4$, that is, 10,000 km horizontal and 10 km vertical. How many grid points would you need, if they have to be spaced out at the scale $L$ you determined in the last problem!

(7) Based on problem 6, we can now ignore the viscous term. We'd now like to compare the nonlinear terms to the Coriolis term, $fv$. In question 5, you showed that the nonlinear terms scale as $U^2/L$.

(a) Recall that $f = 2\Omega \sin \phi$, where $\Omega$ is the angular velocity of Earth, or 1 revolution every day: $2\pi/86400$ radians/s. Assuming $v$ scales as $U \approx 10$ m/s, as we did above, what is the scale of $fv$ at (i) the equator, (ii) 30°N $= \pi/6$ radians, (iii) 45°N $= \pi/4$ radians, and (iv) the pole.

(b) Show that ratio of the nonlinear terms to the Coriolis term is

$$\frac{U}{fL} \equiv Ro$$

where because this ratio is also so important, we gave it a name, the Rossby number $Ro$.

(c) What is the Rossby number at the latitudes you considered in part (a), namely (i) the equator, (ii) 30°N $= \pi/6$ radians, (iii) 45°N $= \pi/4$ radians, and (iv) the pole. Remember that we have chosen a length scale $L = 10^6$ m (1000 km).

(d) When the Rossby number is small, the nonlinear terms are dwarfed by the Coriolis term, and we can vote them off the island, too! For what latitudes is $Ro$ small?

(8) We are almost to the finish line, but what do we do with the first term in the momentum equation, $\frac{\partial u}{\partial t}$? We know that $u$ scales as $U$, but how does time $t$ scale? A reasonable way to make progress is to assume that time scales as $T$, the time it takes the a flow with speed $U$ to cross the characteristic length scale $L$, or

$$T \approx \frac{L}{U}$$

Using this scaling, show that $\frac{\partial u}{\partial t}$ scales the same as the nonlinear terms: $U^2/L$. This means that when the Rossby number is small, we can ignore the time derivative, too!

(9) From the big nasty nonlinear PDE that we started with, we are left with just:

$$fv = -\frac{1}{\rho}\frac{\partial p}{\partial x}$$

and if we did the same work for the meridional momentum equation, we would get

$$-fu = -\frac{1}{\rho}\frac{\partial p}{\partial y}$$

(a) What is this balance called?

(b) Suppose there is a low pressure system (as with a nor'easter or typical winter storm). Sketch how the winds would blow about this low pressure.

(c) Suppose there is a high pressure system (as when there is a heat wave). Sketch how the winds would blow about this high pressure.

(10) A major winter storm can have a central pressure as low as 970 hPa and radius of 1000 km: $l = 1 \cdot 10^6$ m. We will very, very crudely assume that the pressure is

$$p(x, y) = 10^5 - 3 \times 10^3 e^{-\frac{x^2+y^2}{l^2}},$$

where we are using SI units of Pa. Here, the storm is centered around the point $(x, y) = (0, 0)$ where the pressure is $9.7 \cdot 10^4$ Pa. Assume that the storm is centered at 45 degrees, so $f = 2\Omega \sin(\pi/4)$. You are at the surface, so the density of air is approximately 1.2 kg/m$^3$

(a) Using geostrophic balance, what would the meridional wind velocity $v(x)$ be along the line $y = 0$. (You would get a similar profile in other directions.)

(b) What is the maximum meridional wind? You can plot your function or use calculus to solve this. Note that by symmetry you will get this value and it's opposite on different sides of the storm.

(11) Now let's try something more exotic. A major hurricane can have a central pressure of 900 hPa. Hurricanes are also much smaller than winter storms, with radius of just 30 km: $l = 3 \cdot 10^4$ m. Crudely assume that the pressure is

$$p(x, y) = 10^5 - 10^4 e^{-\frac{x^2+y^2}{l^2}}.$$

As with the last problem, the hurricane is centered around the point $(x, y) = (0, 0)$. A massive storm like this will be in the tropics, so assume it is at 30 N: $f = 2\Omega \sin(\pi/6)$.

(a) Following the last problem, use geostrophic balance to compute the meridional wind velocity $v(x)$ be along the line $y = 0$. (You shouldn't have to any more work from the last problem, other than changing the constants!)

(b) What is the maximum velocity now? If you have done this correctly, you will find that this velocity is excessively large. Now we really need that scram jet!

(c) At this point, you should be concerned that your professor has messed something up. To help save the day, check if the assumptions you had to make to get geostrophic balance are valid. Recall that in problem 7b, the key to eliminating the nonlinear terms is the Rossby number $Ro$. For the high winds we are predicting now, what is the Rossby number?

(d) It turns out that indeed hurricanes are not low Rossby number flows. Rather, you can re-write the momentum equations in cylindrical coordinates around the storm center and find that the appropriate balance is "cyclostrophic balance", where the pressure gradient balances the centrifugal forces that would tear the storm apart. Let $u$ be the tangential wind around the storm and $r$ your distance from the center. The balance is:

$$\frac{u^2}{r} \approx -\frac{1}{\rho}\frac{\partial p}{\partial r}$$

Assume the pressure $p(r) = 10^5 - 10^4 e^{-\frac{r^2}{l^2}}$ and compute the maximum wind.

(e) Your answer should be much more reasonable, though still dangerously high: this is a monster storm. Confirm that the Rossby number is not small, justifying the use of cyclostrophic balance.

(12) Finally, use the thermal wind relationship to explain why there are westerly jets in both hemispheres. Please be quantitative, relating an estimate of the equator-to-pole temperature gradient to the strength of the jets.